

Movement in and out
of cells

Diffusion: The net movement of particles from a region of high concentration gradient to a region of low concentration down the concentration gradient due to the random movement of particles.

- * Diffusion takes place passively without energy
- * Does not require a partially permeable membrane

What affects diffusion rate?

1- **Surface area:** Increased by the presence of cilia, villi, microvilli: Hair-like structures that increase surface area.

2- **Temperature:** As the temperature increases, particles gain more kinetic energy, so diffuse faster

3- **Concentration gradient:** The greater the difference in the concentration gradient of a substance on either side of a membrane, the faster it will diffuse.

4- **Distance travelled:** The thinner the cell is, the less distance is travelled, so the faster the diffusion rate

Osmosis: The net movement of water molecules from a region of higher water potential to a region of lower water potential down the water potential gradient through a partially permeable membrane.

- * Passive process, so doesn't require energy
- * It requires a partially permeable membrane like that of a visking tube or a cell membrane.

Osmosis in animal cells

- ① Hypotonic solution: If an animal cell is placed in pure (distilled) water, water will move by osmosis from high water potential (Solution) to low water potential (cell) down the water potential gradient through a partially permeable membrane. So, the cell gains water, and swells up; it will eventually burst.
- ② Hypertonic solution: If an animal cell is placed in a concentrated sugar/salt solution, water will move from high water potential (cell) to low water potential (solution) down the water potential gradient through a partially permeable membrane. So the cell loses water, and decreases in size until it shrivels / shrinks.
- ③ Isotonic solution: If the cell is placed in an isotonic solution, there will be no net movement of water molecules since water will move in equal rates into and out of the cell as water potential is equal in and out of the cell. So no osmosis occurs and the shape and size of the cell stay the same.

Osmosis in Plant cells

① Hypotonic solution: If a plant cell is placed in pure (distilled) water, water will move by osmosis from high water potential (Solution) to low water potential (cell) down the water potential gradient through a partially permeable membrane. So, the cell gains water and swells. The cytoplasm and vacuole increase in size. The cell becomes turgid because the cell wall prevents it from bursting.

② Hypertonic solution: If a plant cell is placed in a concentrated sugar/salt solution, water will move from high water potential (cell) to low water potential (solution) down the water potential gradient through a partially permeable membrane. So the cell loses water, and the vacuole and the cytoplasm decrease in size until it becomes flaccid. (Hypertonic $\rightarrow$ isotonic)

③ If a cell is placed in a VERY concentrated sugar/salt solution, water moves from high water potential (cell) to low water potential down the water potential gradient through a partially permeable membrane. The cell loses water, and the vacuole and cytoplasm decrease in size up until the cell membrane detaches from the cell wall, and the cell becomes plasmolysed.

④ Isotonic solution: If the cell is placed in an isotonic solution, there will be no net movement of water molecules since water will move in equal rates into and out of the cell as water potential is equal in and out of the cell. So no osmosis occurs and the shape and size of the cell stay the same.

Active transport: The movement of molecules or ions from a region of low concentration to a region of high concentration against the concentration gradient through the cell membrane using energy from respiration as well as a carrier and channel proteins.

* **Protein carrier** can change shape to engulf molecules and shuttle them across the cell membrane.

* **Channel proteins:** Tunnels in the cell membrane that open to allow certain molecules to cross the membrane

Importance of active transport:

- 1- Allows molecules to be absorbed from dilute solutions
- 2- Harmful molecules can be removed from cells against concentration gradient.